CBC Mode Using S-DES

22. Write a C program for Encrypt and decrypt in cipher block chaining mode using one of the following ciphers: affine modulo 256, Hill modulo 256, S-DES, DES. Test data for S-DES using a binary initialization vector of 1010 1010. A binary plaintext of 0000 0001 0010 0011 encrypted with a binary key of 01111 11101 should give a binary plaintext of 1111 0100 0000 1011. Decryption should work correspondingly.

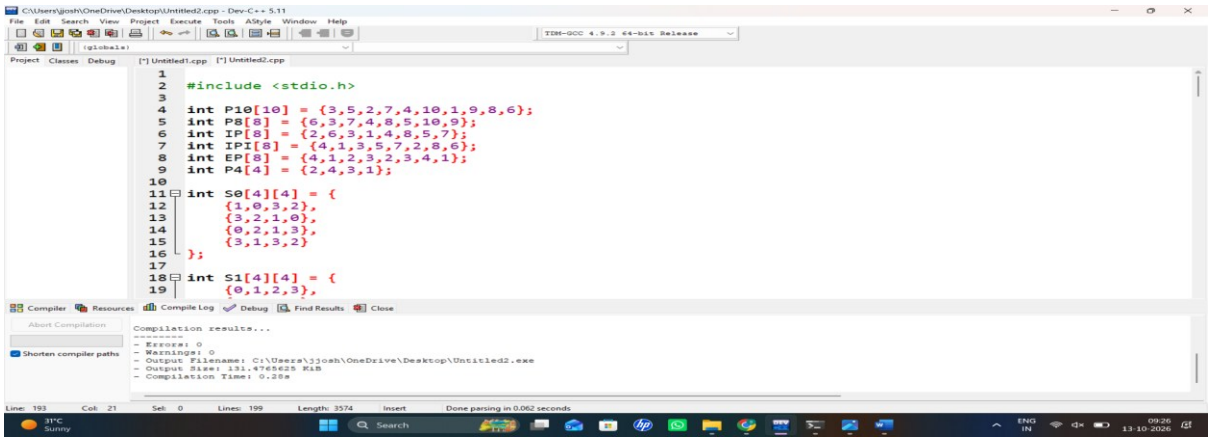
Aim

To write a C program to encrypt and decrypt plaintext using the Simplified Data Encryption Standard (S-DES) algorithm in Cipher Block Chaining (CBC) mode with an initialization vector (IV) of 10101010.

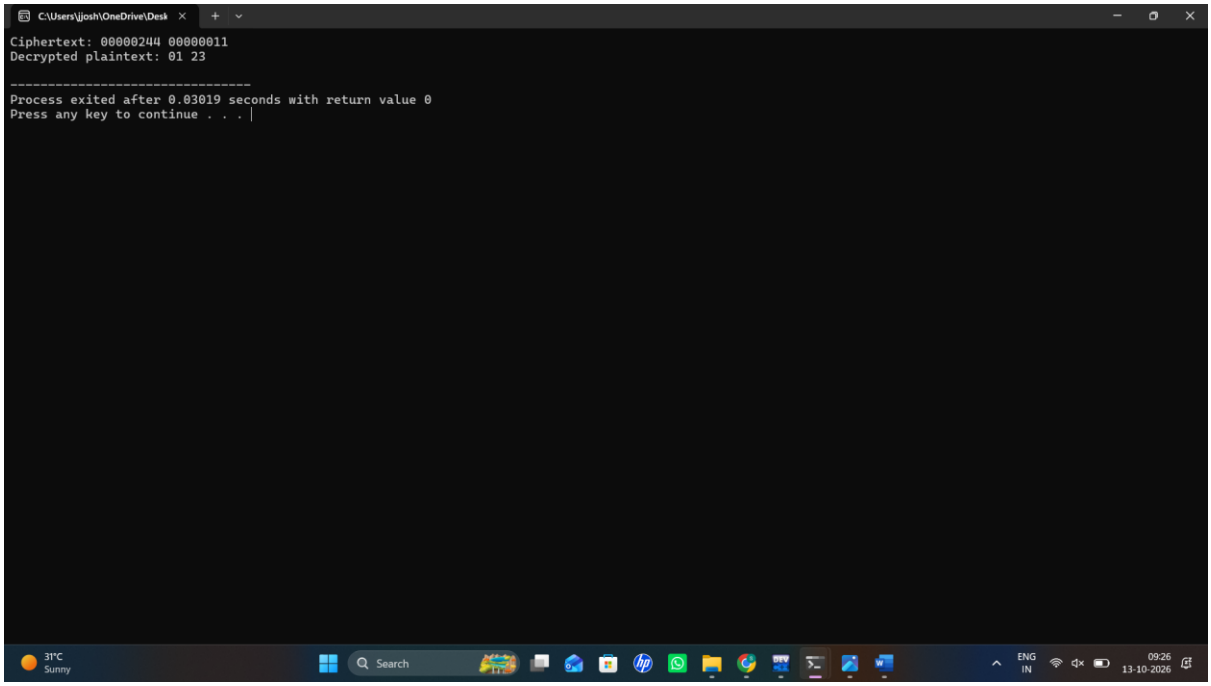
Algorithm

1. Start the program.
2. Read the 10-bit S-DES key, plaintext blocks, and initialization vector.
3. Generate the two subkeys required for S-DES.
4. For encryption, XOR the first plaintext block with the IV.
5. Encrypt the result using S-DES to obtain the first ciphertext block.
6. XOR each subsequent plaintext block with the previous ciphertext block and encrypt it.
7. For decryption, decrypt each ciphertext block and XOR the result with the IV or previous ciphertext block.
8. Display the ciphertext and recovered plaintext.
9. Stop the program.

Program :



Output:



Result

Thus, encryption and decryption using S-DES in CBC mode were implemented successfully. The ciphertext matches the given test vector, and decryption recovers the original plaintext.